POSTNIKOV UNIQUENESS GIVES SIBLING AGREEMENT

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Postnikov systems. derived-example-key-postnikov, derived-lemma-existence-postnikov-system (cross-chapter, named). Construction and uniqueness of Postnikov systems.

Source reference. Original reference 0D8Z: focused-seeds/geometry/S9-eqv-sibling-fully-faithful/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0D83: focused-seeds/geometry/S9-eqv-sibling-fully-faithful/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{D}$ be a triangulated category. Let $F, F' : D^b(\mathcal{A}) \longrightarrow \mathcal{D}$ be exact functors of triangulated categories. Assume*

1. *the functors $F \circ i$ and $F' \circ i$ are isomorphic where $i : \mathcal{A} \to D^b(\mathcal{A})$ is the inclusion functor, and*
2. *for all $X, Y \in \mathrm{Ob}(\mathcal{A})$ we have $\mathrm{Ext}^q_{\mathcal{D}}(F(X), F(Y)) = 0$ for $q < 0$ (for example if F is fully faithful).*

Then F and F' are siblings.

Proof. Let $K \in D^b(\mathcal{A})$. We will show $F(K)$ is isomorphic to $F'(K)$. We can represent K by a bounded complex $A^\bullet$ of objects of $\mathcal{A}$. After replacing K by a translation we may assume $A^i = 0$ for $i > 0$. Choose $n \geq 0$ such that $A^{-i} = 0$ for $i > n$. The objects

$$M_i = (A^{-i} \to \ldots \to A^0)[-i], \quad i = 0, \ldots, n$$

form a Postnikov system in $D^b(\mathcal{A})$ for the complex $A^\bullet = A^{-n} \to \ldots \to A^0$ in $D^b(\mathcal{A})$. See Derived Categories, Example 0D8Z. Since both F and F' are exact functors of triangulated categories both

$$F(M_i) \quad \text{and} \quad F'(M_i)$$

form a Postnikov system in $\mathcal{D}$ for the complex

$$F(A^{-n}) \to \ldots \to F(A^0) = F'(A^{-n}) \to \ldots \to F'(A^0)$$

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Since all negative Exts between these objects vanish by assumption we conclude by uniqueness of Postnikov systems (Derived Categories, Lemma 0D83) that $F(K) = F(M_n[n]) \cong F'(M_n[n]) = F'(K)$. $\square$